

Mohankumar Markuli Chandrayigowda

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PROFESSIONAL SUMMARY

AI Engineer with expertise in building intelligent backend systems and Retrieval-Augmented Generation pipelines using Python, LangChain, and LLM APIs. Four plus years of production experience integrating machine learning concepts into scalable applications serving thirty thousand plus concurrent users. Skilled in prompt engineering, vector database design, semantic search, and grounding LLM responses with domain-specific knowledge. Passionate about ethical AI implementation and practical applications of generative AI in production systems.

EDUCATION

University of Liverpool

Master of Science in Business Analytics and Big Data; CGPA: 6.7

Liverpool, United Kingdom

Sep 2022 – Dec 2023

Rajeev Institute of Technology

Bachelor of Engineering in Computer Science; CGPA: 7.41

Hassan, India

Aug 2016 – Sep 2020

EXPERIENCE

Freelance - Hospitality and Retail Industry

Business Analyst Consultant

Liverpool, United Kingdom

Feb 2024 – Jan 2026

- Developed **Python**-based data pipeline analyzing worker performance across 15+ hospitality and logistics roles, processing 500+ active freelance workers through **exploratory data analysis** and **regression modeling** to predict reliability metrics.
- Designed transparent per-shift compensation **strategy system** with base and holiday pay differentials, validated using Python data quality checks on existing schema, **increasing freelancer platform adoption** by 3x and **reducing worker churn** versus competing agencies pay rates.

Infosys Limited

Systems Engineer

Bengaluru, India

Dec 2020 – Jul 2022

- Redesigned static web application to fully **responsive design using Angular and Bootstrap** MEAN stack product, expanding mobile accessibility for 500+ trainees during 3 month onboarding cycle for **GenQ assessment platform**.
- Analyzed learner performance data using **pandas**, and **numPy** across 23K to 27K daily active users, identifying top performers and reducing talent acquisition team screening time by 80% through automated candidate filtering for **Springboard EdTech Product**.
- Supported infrastructure for **Zoe chatbot microservice** integrated within Springboard, ensuring 0.999 uptime across enterprise services.
- Delivered **200+ plus training and sales sessions** on **AI, machine learning, big data**, and **Spark** to senior executives, team leads and on-boarding engineers, accelerating team ramp-up.

Infosys Limited

Software Engineering Internship – Spring Boot

Mysore, India

Feb 2020 – Mar 2020

- Built **REST APIs in Java and Spring Boot** handling 72+ product SKUs and orders, with **MySQL** queries optimized to sub-second retrieval times for Wintership ecommerce project.
- Structured layered architecture using **controller-service-repository** pattern, reducing code duplication by forty percent and enabling seamless **API testing** across 18 endpoints via **Postman**.

Adamos Technologies

Internship – Web Technologies and IOT

Mysore, India

Jul 2019 – Aug 2019

- Completed hands-on **web development** and **IoT** training covering 8 embedded systems modules across **Arduino** and **Raspberry Pi**, mastering microcontroller programming through practical exercises that directly enabled capstone project delivery.

- Designed and built proof-of-concept **advanced traffic management system** using **Raspberry Pi**, **IR sensors**, and **Python** for a 3 way and 4 way junction model, dynamically adjusting traffic signal timing based on vehicle density detection with real-time **data logging to Firebase cloud**.

CERTIFICATIONS

- **Airtribe AI first Software Engineer Program** (in Progress), Comprehensive backend engineering curriculum covering JavaScript fundamentals, REST APIs, asynchronous programming, microservices, JWT authentication, unit and integration testing, CI/CD with GitHub Actions, and emerging modules in RAG systems, LLM integration, and relational databases.
- **Great Learning Executive Program in Artificial Intelligence and Machine Learning** (In partnership with Infosys), November 2022. Covered machine learning fundamentals, data analytics, exploratory data analysis, data preprocessing, handling missing values, and decision-making with data.

PROJECTS

- **PDF Loader RAG Pipeline** — Python, LangChain, OpenAI API or Ollama, Built an end-to-end Retrieval-Augmented Generation pipeline that loads PDF documents, chunks and embeds content into a vector store, and retrieves semantically relevant context to ground LLM responses, reducing hallucinations and enabling accurate document-based question answering.
- **Ethical Issues with the use of AI in Business Operations** — MSc Dissertation, University of Liverpool, December 2023. Research-driven analysis addressing three critical ethical dimensions: job displacement from automation, accountability when AI systems fail, and jurisdictional conflicts across different country regulations. Published on Medium.
- **News Aggregator API** — Node.js, Express, MongoDB, JWT, GNews API. Built personalized news aggregation backend with JWT authentication, user preference management, in-memory caching with background refresh to reduce external API calls, and full unit and integration test coverage.
- **FinEdge** Personal Finance & Expense Tracker API — Node.js, Express, MongoDB, JWT. Developed personal finance API with transaction CRUD, advanced filtering across five parameters, and MongoDB aggregation pipelines delivering monthly and yearly analytics with pagination support.
- **devTinder** Developer Networking Platform Backend — Node.js, Express, MongoDB. Built backend for a developer matchmaking platform featuring JWT authentication, profile management, and bi-directional connection request system with four interaction states enabling dynamic user feed generation.